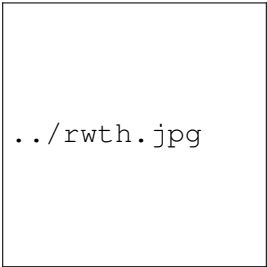


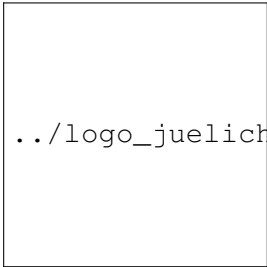
Measurement of the deuteron static and oscillating electric dipole moment at the COoler SYnchrotron COSY

J. Pretz

RWTH Aachen & FZ Jülich
on behalf of the JEDI collaboration



../rwth.jpg



../logo_juelich_2018.jpg

COoler SYnchrotron COSY

- storage ring at Forschungszentrum Jülich in Germany
- polarized proton & deuteron beam up to a momentum of 3.7 GeV/c
- JEDI (Jülich Electric Dipole moment Investigations) collaboration
- stopped operation in 2023

../Cosy_Foto.jpg

../logoJEDI_nobgL.png

SSP 2018 Aachen, Germany → SSP 2025 Nara

../hapof_jul_2024/ac004.jpg

Aachen Cathedral



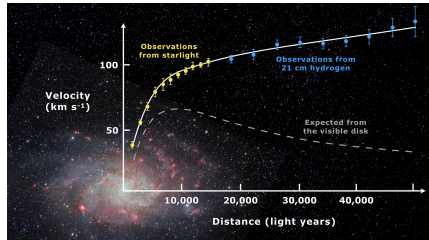
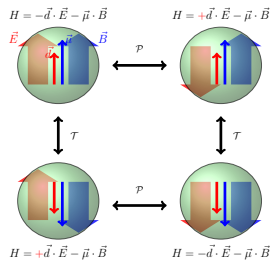
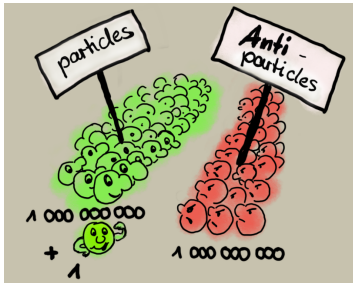
Todai-ji temple

both conference venues in walking distance from UNESCO world heritage site,
built at the same time

Motivation

Standard Model of Particle Physics successful but . . .

- Fails to explain matter-antimatter asymmetry in the universe
- Why is CP-violation in the strong sector not present (although allowed)?
- What does Dark Matter consists of?



source: M. De Leo, Wikipedia

Outline

- **Introduction:**

Electric Dipole Moments and connection to axions/ALPs

- **Experimental Methods for charged particles**

Observing and controlling spin precession

- **Experiments & Results:**

on permanent & oscillating electric dipole moments

Introduction

$\mathcal{T}$ and $\mathcal{P}$ violation of EDM

$\vec{d}$: EDM

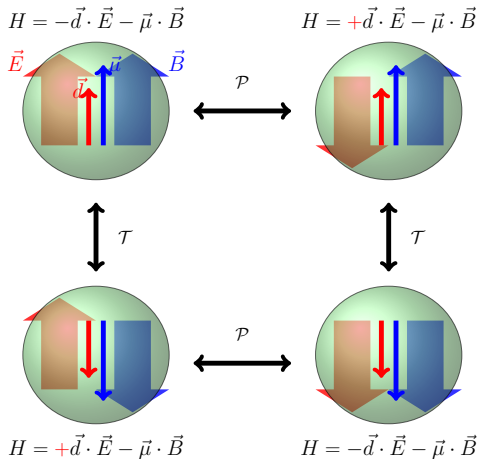
$\vec{\mu}$: magnetic moment (MDM)

both $\parallel$ to spin $\vec{s}$

$$H = -\mu \frac{\vec{s}}{s} \cdot \vec{B} - d \frac{\vec{s}}{s} \cdot \vec{E}$$

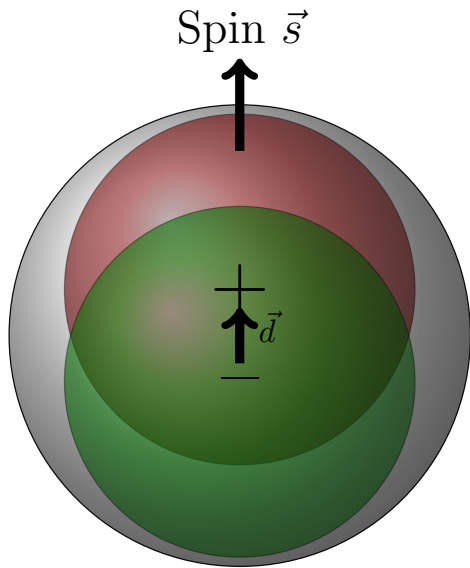
$$\mathcal{T}: H = -\mu \frac{\vec{s}}{s} \cdot \vec{B} + d \frac{\vec{s}}{s} \cdot \vec{E}$$

$$\mathcal{P}: H = -\mu \frac{\vec{s}}{s} \cdot \vec{B} + d \frac{\vec{s}}{s} \cdot \vec{E}$$



$\Rightarrow$ EDM measurement tests violation of fundamental symmetries $\mathcal{P}$ and $\mathcal{T}(\overset{CP\mathcal{T}}{=} C\mathcal{P})$

Electric Dipole Moments (EDM)



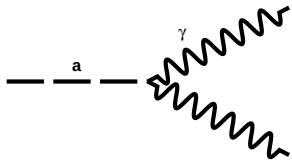
- permanent separation of positive and negative charge
- fundamental property of particles (like magnetic moment, mass, charge)
- existence of EDM only possible via violation of time reversal $\mathcal{T} \stackrel{CP\mathcal{T}}{=} \mathcal{CP}$ and parity $\mathcal{P}$ symmetry
- close connection to “matter-antimatter” asymmetry
- axion field leads to oscillating EDM
$$d = d_{DC} + d_{AC} \cos(\omega_a t + \varphi_a)$$
$$m_a c^2 = \hbar \omega_a$$

Axions/Axion Like Particles (ALPs)

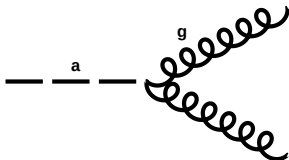
- hypothetical pseudo-scalar elementary particle postulated by Peccei,Quinn,Wilczek,Weinberg to resolve the strong CP problem
- axion are also Dark Matter candidates
- axion like particles (ALP): similar properties as axions, (but ALPs don't solve the strong QCD problem)
- huge experimental effort to search for axion/ALPs (haloscopes, helioscopes, light shining through the wall, mainly coupling to photons)
- in storage rings with polarized beams axion-gluon/nucleon coupling can be studied

Axion Coupling

$$\mathcal{L} : -\frac{\alpha}{8\pi} \frac{C_\gamma}{f_a} \mathbf{a} F_{\mu\nu} \tilde{F}^{\mu\nu}$$

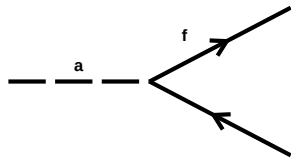


$$-\frac{\alpha_s}{8\pi} \frac{C_G}{f_a} \mathbf{a} G_{\mu\nu}^b \tilde{G}^{b,\mu\nu}$$



oscillating
Electric Dipole Moment (oEDM)

$$-\frac{1}{2} \frac{C_N}{f_a} \partial_\mu \mathbf{a} \bar{\Psi}_f \gamma^\mu \gamma^5 \Psi_f$$

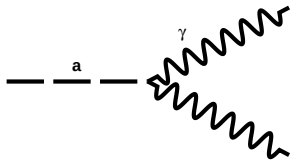


axion wind term

For low axion masses, if axions saturate dark matter they can be described by classical field: $\mathbf{a}(t) = a_0 \cos(\omega_a t + \varphi_a)$, $m_a c^2 = \hbar \omega_a$, Coupling $\propto \frac{1}{f_a} \propto m_a$ [1]

Axion Coupling

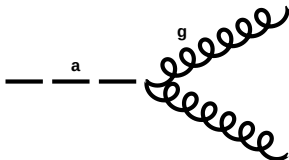
$$\mathcal{L} : -\frac{\alpha}{8\pi} \frac{C_\gamma}{f_a} \mathbf{a} F_{\mu\nu} \tilde{F}^{\mu\nu}$$



studied by many
experiments

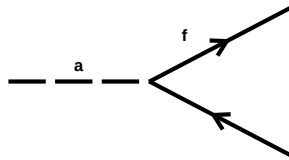
[1]

$$-\frac{\alpha_s}{8\pi} \frac{C_G}{f_a} \mathbf{a} G_{\mu\nu}^b \tilde{G}^{b,\mu\nu}$$



oscillating
Electric Dipole Moment (oEDM)

$$-\frac{1}{2} \frac{C_N}{f_a} \partial_\mu \mathbf{a} \bar{\Psi}_f \gamma^\mu \gamma^5 \Psi_f$$



axion wind term

**accessible in storage ring experiments
with spin polarized beams**

Experimental Methods

Experimental Method

Observe Spin Precession in electric and magnetic fields:

$$\vec{\Omega} = \frac{-d\vec{E} - \mu\vec{B}}{|\vec{s}|}, \quad \dot{\vec{s}} = \vec{\Omega} \times \vec{s}$$

Order of magnitude:

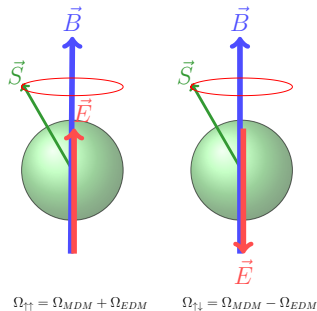
Neutron in earth B -field:

$$\Omega_{\text{MDM}} \approx 9000 \text{ s}^{-1}$$

$$d_n = 1 \times 10^{-26} \text{ e} \cdot \text{cm}$$

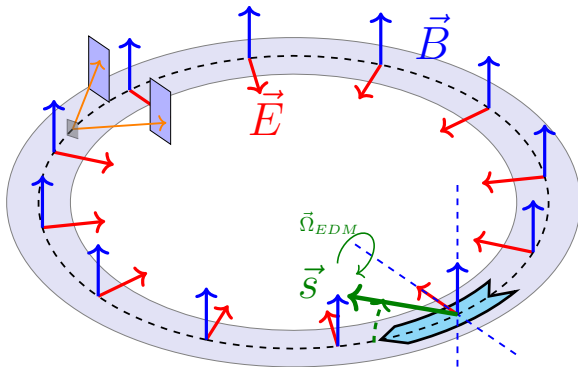
in electric field $E = 10^7 \text{ V/m}$:

$$\Omega_{\text{EDM}} \approx 3 \times 10^{-6} \text{ s}^{-1}$$



Even more complicated for charged particles:

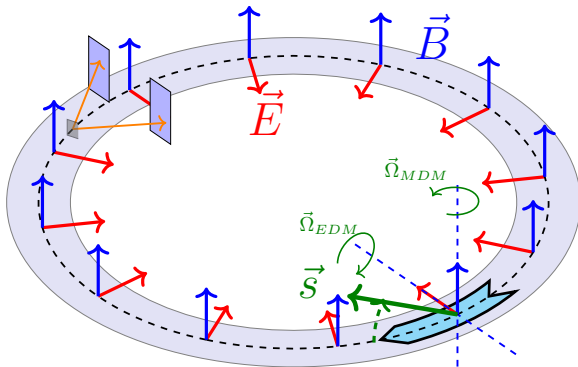
Experimental Method for charged particle: Storage Ring



$$\frac{d\vec{s}}{dt} \propto \underbrace{d(\vec{E} + \vec{v} \times \vec{B})}_{= \vec{\Omega}_{EDM}} \times \vec{s}$$

build-up of vertical polarization $s_{\perp} \propto d$, if $\vec{s}_{horz} || \vec{p}$ (**frozen spin**)

Experimental Method for charged particle: Storage Ring



$$\frac{d\vec{s}}{dt} \propto \underbrace{d(\vec{E} + \vec{v} \times \vec{B})}_{= \vec{\Omega}_{EDM}} \times \vec{s}$$

In general:

$$\frac{d\vec{s}}{dt} = (\vec{\Omega}_{MDM} + \vec{\Omega}_{EDM}) \times \vec{s}$$

build-up of vertical polarization $s_{\perp} \propto d$, if $\vec{s}_{horz} || \vec{p}$ (**frozen spin**)

Spin Precession: Thomas-BMT Equation

$$\frac{d\vec{s}}{dt} = \vec{\Omega} \times \vec{s} = \frac{-q}{m} \left[\underbrace{G\vec{B} + \left(G - \frac{1}{\gamma^2 - 1}\right) \vec{v} \times \vec{E}}_{= \vec{\Omega}_{\text{MDM}}} + \underbrace{\frac{\eta}{2c}(\vec{E} + \vec{v} \times \vec{B})}_{= \vec{\Omega}_{\text{EDM}}} \right] \times \vec{s}$$

electric dipole moment (EDM): $\vec{d} = \eta \frac{q\hbar}{2mc} \vec{s}$,

magnetic dipole moment (MDM): $\vec{\mu} = 2(G + 1) \frac{q\hbar}{2m} \vec{s}$

Note: $\eta = 2 \cdot 10^{-15}$ for $d = 10^{-29}$ ecm, $G \approx 1.79$ (-0.14) for protons (deuterons)

Spin Precession: Thomas-BMT Equation

$$\frac{d\vec{s}}{dt} = \vec{\Omega} \times \vec{s} = \frac{-q}{m} \left[\underbrace{G\vec{B} + \left(G - \frac{1}{\gamma^2 - 1}\right) \vec{v} \times \vec{E}}_{\vec{\Omega}_{\text{MDM}} = 0, \text{ frozen spin}} + \underbrace{\frac{\eta}{2c}(\vec{E} + \vec{v} \times \vec{B})}_{= \vec{\Omega}_{\text{EDM}}} \right] \times \vec{s}$$

frozen spin achievable with pure electric field if $G = \frac{1}{\gamma^2 - 1}$,

works only for $G > 0$, e.g. proton

or with special combination of E , B fields and γ , i.e. momentum

Spin Precession: Thomas-BMT Equation

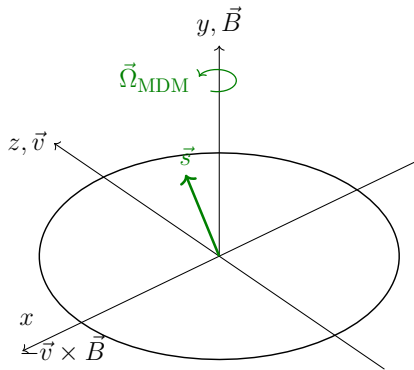
$$\frac{d\vec{s}}{dt} = \vec{\Omega} \times \vec{s} = \frac{-q}{m} \left[\underbrace{G\vec{B}}_{=\vec{\Omega}_{\text{MDM}}} + \left(G - \frac{1}{\gamma^2 - 1} \right) \vec{v} \times \vec{E} + \underbrace{\frac{\eta}{2c}(\vec{E} + \vec{v} \times \vec{B})}_{=\vec{\Omega}_{\text{EDM}}} \right] \times \vec{s}$$

In COSY, only B -field available:

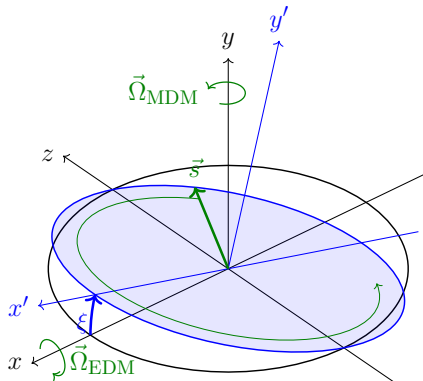
- ⇒ spin precession due to MDM cannot be (completely) suppressed
- ⇒ no macroscopic build-up of vertical polarisation component

Spin Precession

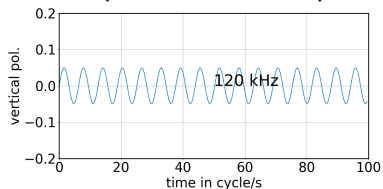
- Without EDM spin rotates around vertical y -axis



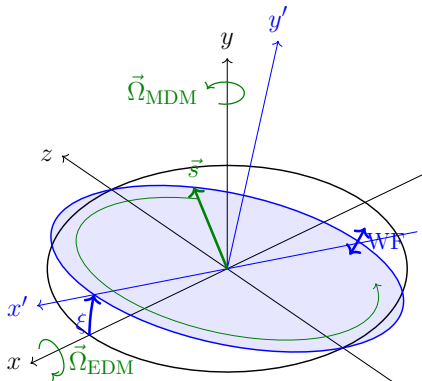
Spin Precession



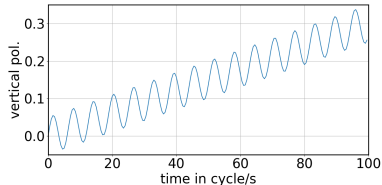
- Without EDM spin rotates around vertical y -axis
- EDM causes a tilt of invariant spin axis in radial x -direction of $\xi \approx \frac{\eta\beta}{2G}$
- This results in an oscillation of the vertical polarisation component



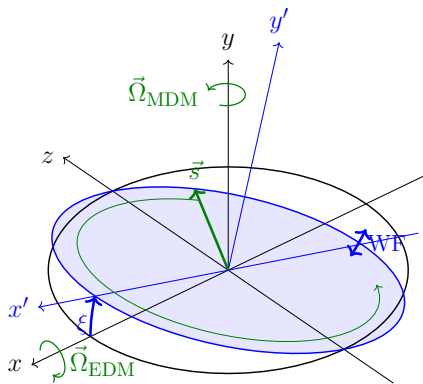
Spin Precession



- Without EDM spin rotates around vertical y -axis
- EDM causes a tilt of invariant spin axis in radial x -direction of $\xi \approx \frac{\eta\beta}{2G}$
- This results in an oscillation of the vertical polarisation component
- With a resonant spin kick (provided by a rf Wien filter) a macroscopic build-up of a vertical polarisation can be achieved



Spin Precession



- Without EDM spin rotates around vertical y -axis
- EDM causes a tilt of invariant spin axis in radial x -direction of $\xi \approx \frac{\eta\beta}{2G}$
- This results in an oscillation of the vertical polarisation component
- With an resonant spin kick (provided by a rf Wien filter) a macroscopic build-up of a vertical polarisation can be achieved
- Misalignments, ... lead to additional tilt in precession plane in x and z -direction

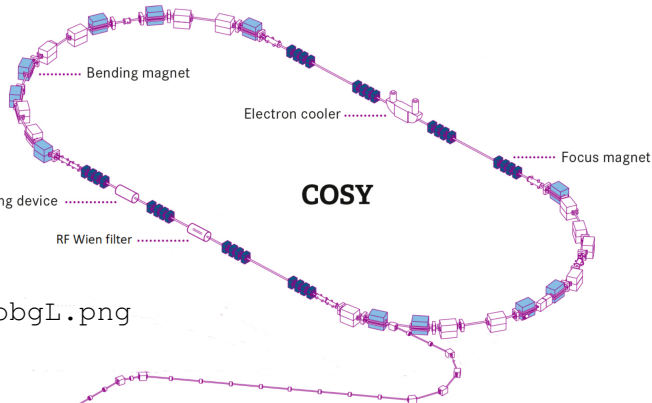
Experiments & Results

Experiments at COSY

COSY circumference	183 m
deuteron momentum	0.970 GeV/c
$\beta(\gamma)$	0.459 (1.126)
magnetic anomaly G	≈ -0.143
revolution frequency f_{rev}	752543 Hz
cycle length	100-1500 s
nb. of stored particles/cycle	$\approx 10^9$

../logoJEDI_nobgL.png

JEDI collaboration,



COler SYnchrotron COSY

Tools developed to manipulate and measure beam polarization:

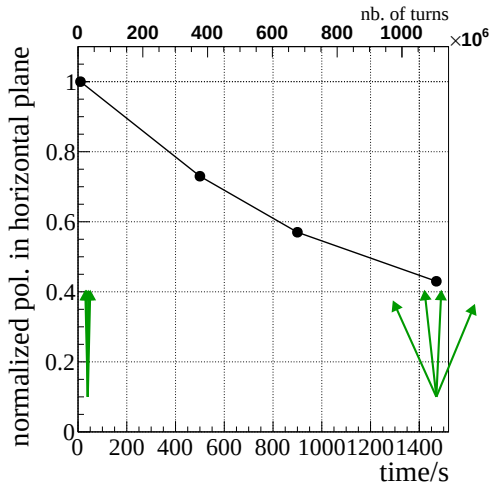
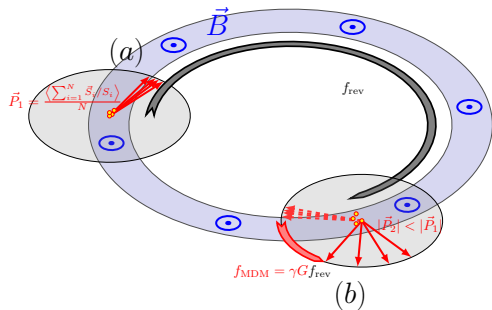
- reaching > 1000 s spin coherence time
- measure 120 kHz spin tune precession in horizontal plane to 10^{-10} in 100 s
- development of polarization feed back system
- radio frequency Wien filter
- (single bunch spin manipulation [3], Beam Position Monitors (BPMs) [4], deflector [5], polarimeter [6], ...)

Results on:

- permanent deuteron EDM
- oscillating deuteron EDM

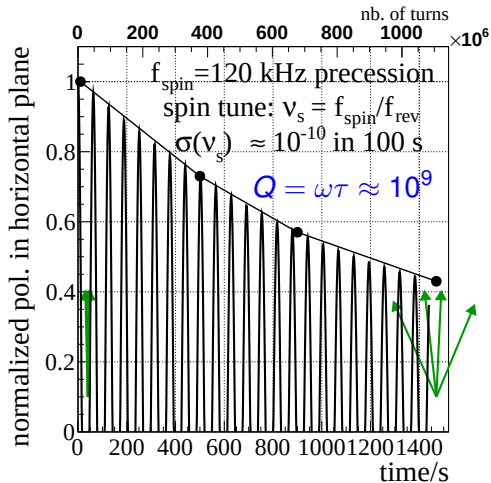
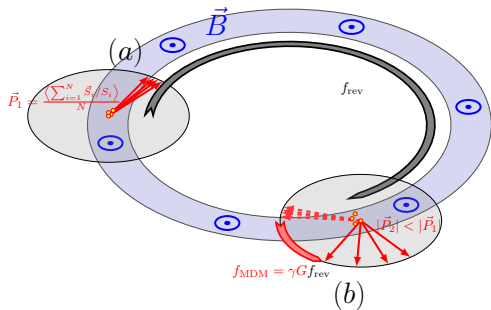
Long Spin Coherence Time (SCT)

Long Spin Coherence time > 1000 s reached

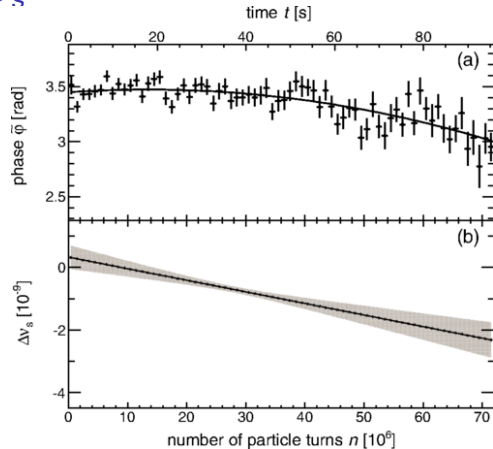
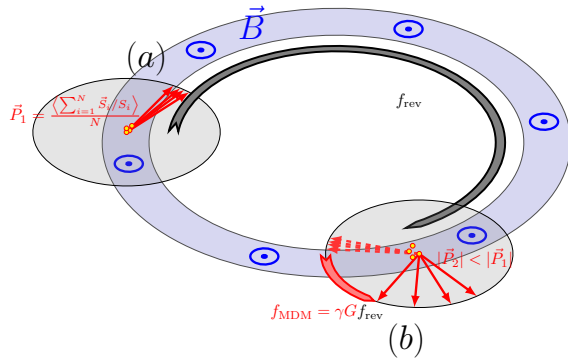


Long Spin Coherence Time (SCT)

Long Spin Coherence time > 1000 s reached



Spin Tune ν_s



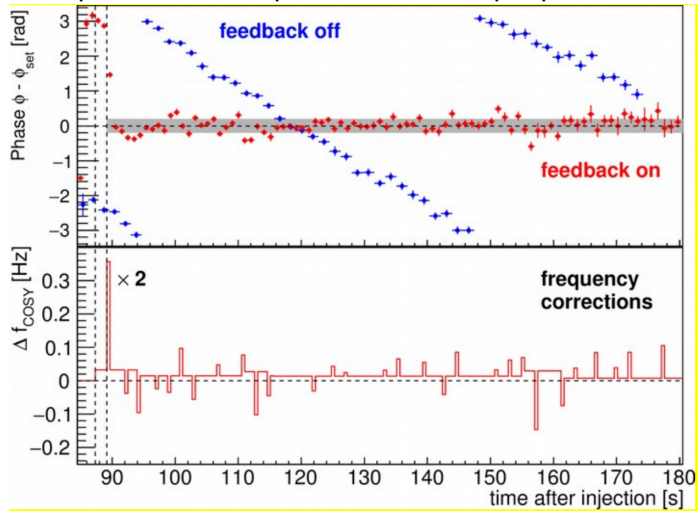
$$\text{Spin tune } \nu_s = \frac{\Omega_{MDM}}{\Omega_{rev}} \approx \gamma G$$

$$\sigma(\nu_s = \gamma G) \approx 10^{-10} \text{ in } 100 \text{ s}$$

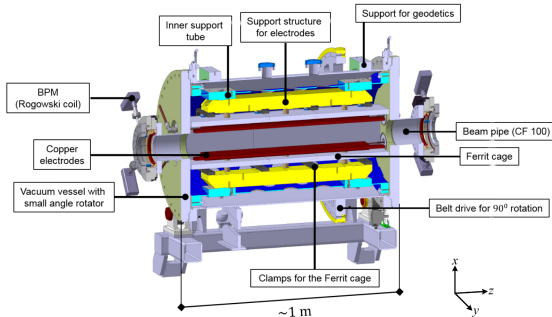
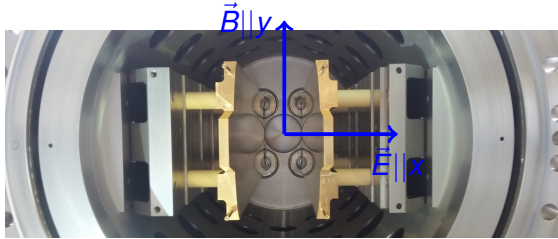
$$\sigma(\nu_s = \gamma G) \approx 10^{-8} \text{ in } 2 \text{ s}$$

Polarisation feedback

Controlling 120kHz precession, Φ : phase between spin precession and rf-device



Radio frequency Wien filter



- field: $2.7 \cdot 10^{-2} \text{Tmm}$ for 1kW input power
- frequency range: 100 kHz-2MHz
- $\vec{F}_L = e(\vec{E} + \vec{v} \times \vec{B}) = 0$
 $\Rightarrow$ no influence on orbit, but on spin

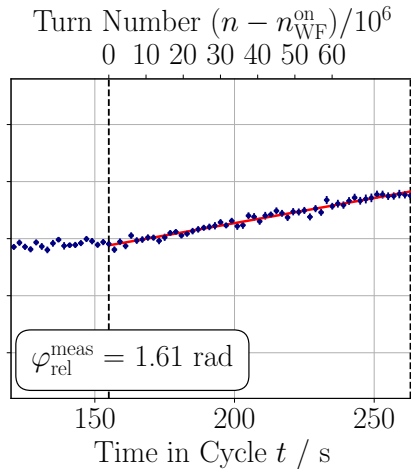
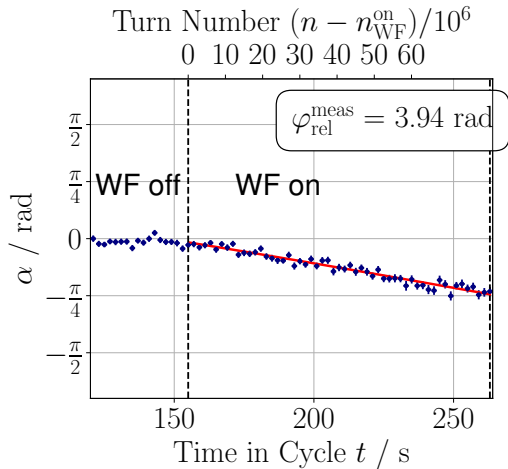
Towards deuteron EDM measurement

Observable: Build-up of vertical polarisation

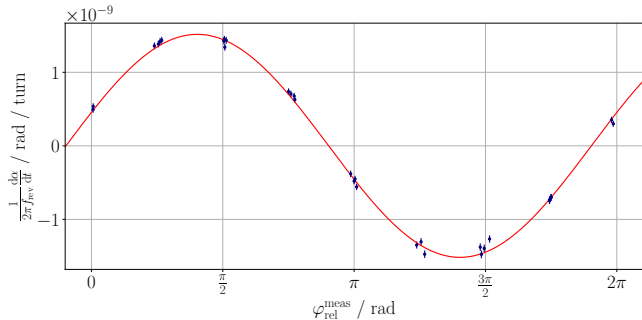
$$\frac{d\alpha}{dt} = \frac{1}{2} \left(\frac{\eta\beta}{2G} + \xi_{syst} \right) \Psi_0 f_{rev} \cos(\varphi_{rel})$$

- $\alpha = \frac{P_V}{P_H}$, ratio of vertical to horizontal polarisation
- Ψ_0 : amplitude of Wien filter spin kick per turn
- φ_{rel} = phase between WF and spin precession
- $\vec{d} = \eta \frac{q\hbar}{2mc} \vec{s}$
- ξ_{syst} : due to misalignments, field imperfections, ...

Polarisation Build-up



Slopes as a function of φ_{rel}



- Measured amplitude would correspond to EDM of d_{deuteron} of $1.6 \approx 10^{-17}$ ecm or to e.g. magnet misalignments of $\mathcal{O}(\text{mrad})$
- Result dominated by systematics. $\Rightarrow$ EDM limit 95% CL for deuteron of $\approx 3 \cdot 10^{-17}$ ecm (preliminary)

Comparison to other direct EDM measurements

Only other direct measurement of charged particle exists for muon

	d	G
BNL muon 95% CL	$1.9 \times 10^{-19} \text{ cm}$	$\approx \frac{\alpha}{2\pi} = 0.001$
JEDI deuteron EDM 95% CL	$\approx 3 \times 10^{-17} \text{ cm}$	-0.14

Note, sensitivity $\propto \frac{d}{G}$

Deuteron result obtained with conventional storage ring, not designed for precision measurements

$\Rightarrow$ Dedicated storage ring (electric bending elements, $\circ$ & $\ominus$ beams) needed

[11]

Axion Searches

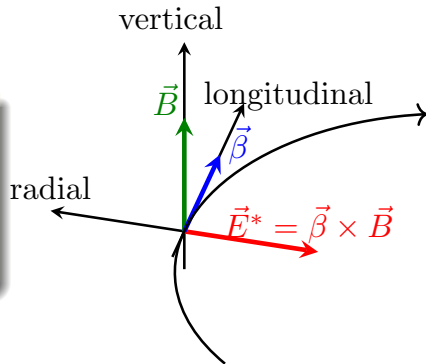
Back to Spin Motion in Storage Ring

$$\frac{d\vec{s}}{dt} = (\vec{\Omega}_{\text{MDM}} + \vec{\Omega}_{\text{EDM}} + \vec{\Omega}_{\text{wind}}) \times \vec{s}$$

$$\vec{\Omega}_{\text{MDM}} = -\frac{q}{m} G \vec{B},$$

$$\vec{\Omega}_{\text{EDM}} = -\frac{1}{S\hbar} d c \vec{\beta} \times \vec{B},$$

$$\vec{\Omega}_{\text{wind}} = -\frac{1}{S\hbar} \frac{C_N}{2f_a} (\hbar \partial_0 \mathbf{a}(t)) \vec{\beta},$$

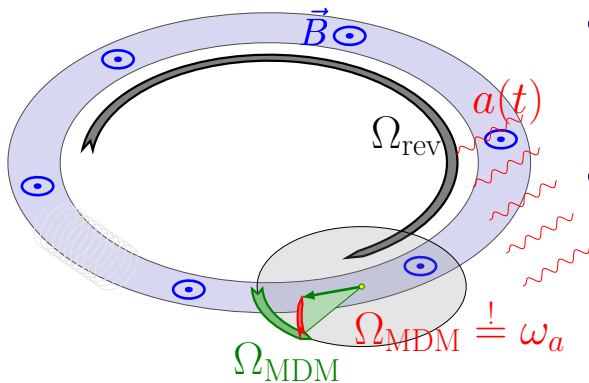


$$d = d_{\text{DC}} + d_{\text{AC}} \cos(\omega_a t + \varphi_0)$$

oscillating EDM: $d_{\text{AC}} = a_0 g_{ad\gamma} \propto C_g$

axion field: $\mathbf{a}(t) = a_0 \cos(\omega_a t + \varphi_0)$

Principle of storage ring axion experiment



- Axion field gives rise to an effective time-dependent θ -QCD term
- This gives rise to an oscillating electric dipole moment EDM d .

$$d = d_{DC} + d_{AC} \sin(\omega_a t + \varphi_a)$$

$$\omega_a = \frac{m_a c^2}{\hbar}$$

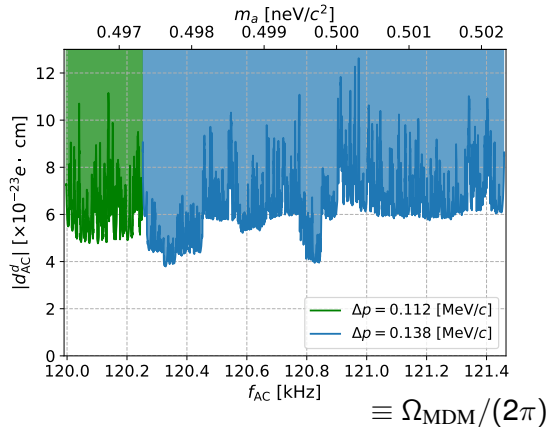
Properties of Method

- AC measurement (i.e. systematics are under control)
- axion wind effect enhanced in storage rings ($v_{\text{particle}} \approx c$)

$$\vec{\Omega}_{\text{wind}} = -\frac{1}{S\hbar} \frac{C_N}{2f_a} (\hbar\partial_0 \mathbf{a}(t)) \vec{\beta}$$

- One can look for ALPs at a given mass given by Ω_{MDM} or scan a certain mass range by varying Ω_{MDM}

Limits on Oscillating EDM d_{AC} , 90% CL



- a few days of beam time
- $\frac{\Omega_{MDM}}{2\pi} = f_{AC} = \frac{1}{2\pi} \frac{m_a c^2}{\hbar} = \gamma G f_{rev}$
- oscillating EDM $< 6 \times 10^{-23} ecm$

published in PRX: [15]

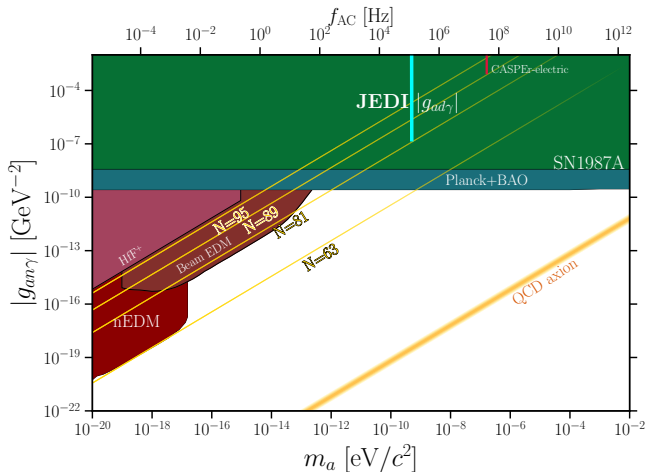
Axion Coupling to EDM operator $g_{ad\gamma}$ (e.g. Axion/Gluon Coupling))

$$g_{ad\gamma} = \frac{d_{AC}}{a_0}$$

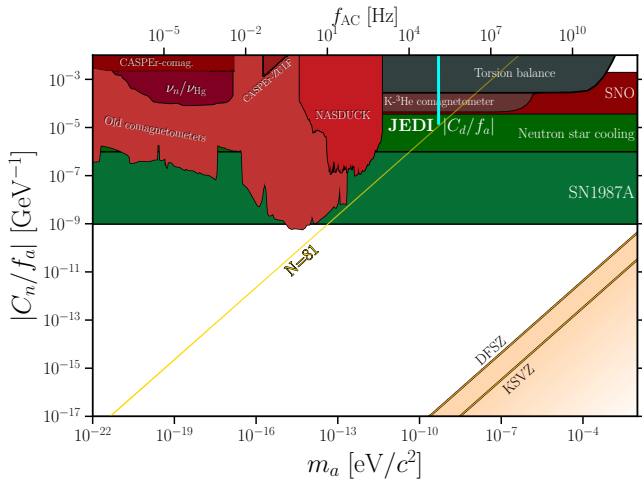
$$a_0 = 0.55 \text{ GeV/cm}^3$$

(Dark Matter is saturated by ALPs)

- assume no axion wind effect
- yellow lines (parallel to QCD axion lines): models with light QCD axion
- JEDI limit comparable or even better compared to other experiments
- Limits from SN1987A, Planck+BAO have strong model dependence



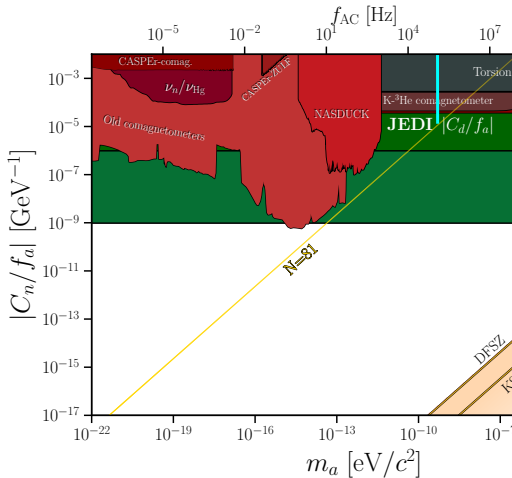
Axion Wind Effect: Coupling to Nucleons C_N/f_a



- storage ring experiments particularly sensitive to axion wind effect ($\beta = \mathcal{O}(1)$)

Axion Wind Effect: Coupling to Nucleons C_N/f_a

2023 PDG:



90. Axions and Other Similar Particles

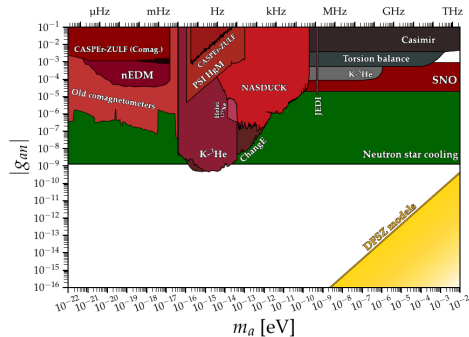
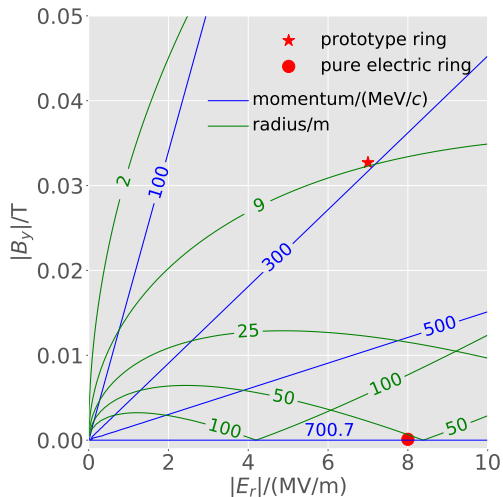


Figure 90.3: Exclusion plot for ALP-neutron coupling as described in the text. Figure courtesy of Ciaran O'Hare [61], includes data from refs. [40, 42, 206, 245–255]. The hadronic axion model prediction is given in Eq. (90.11) with vanishing quark couplings, while the DFSZ model prediction depends on $\tan\beta$ as is found in Eq. (90.12), giving the shaded yellow region above. Note that for a fine-tuned value of $\tan\beta$ g_{an} can be taken to zero. On the other hand, the neutron star cooling constraints [254] also probe the axion-proton coupling g_{ap} at a comparable level (not shown), and both g_{an} and g_{ap} cannot simultaneously be taken to zero in the DFSZ model.

Next Steps

Towards a dedicated storage ring for EDM measurements

Momentum and ring radius for proton in **frozen spin** condition



Two options:

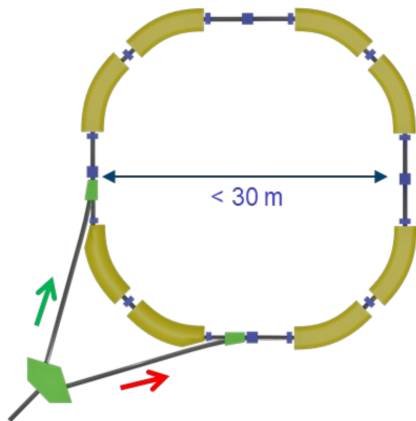
- Pure electric ring:
 $p = 707\text{MeV}$, bending radius ≈ 50 m at $E=8$ MV/m

- ★ combined prototype ring:
 $p = 300\text{MeV}$, bending radius ≈ 9 m at $E=7$ MV/m

Different Options

		
1.) pure magnetic ring	existing (upgraded) COSY ring can be used,	lower sensitivity, precession due to G , i.e. no frozen spin
2.) combined ring	works for $e, p, d, {}^3\text{He}$, smaller ring radius	both $\vec{E}$ and $\vec{B}$ B field reversal for $\odot, \ominus$ required
3.) pure electric ring	no $\vec{B}$ field needed, $\odot, \ominus$ beams simultaneously	works only for particles with $G > 0$ (e.g. e, p)

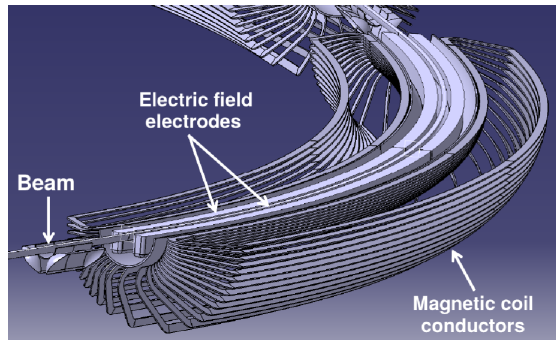
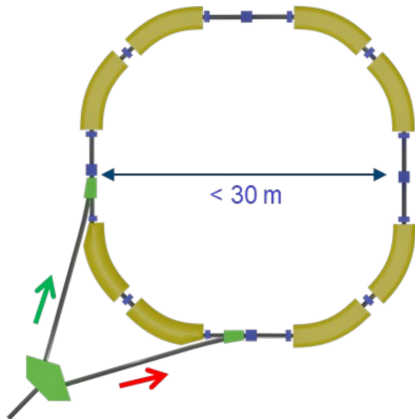
Prototype Ring: Lattice & Bending Element



- operate electrostatic ring
- store $10^9 - 10^{10}$ particles for 1000 s
- simultaneous $\odot$ and $\ominus$ beams
- frozen spin (only possible with additional magnetic bending)
- develop and benchmark simulation tools
- develop key technologies:
beam cooling, deflector, beam position monitors, shielding ...
- perform EDM measurement and axion/ALP search

[16]

Prototype Ring: Lattice & Bending Element



Summary & Outlook

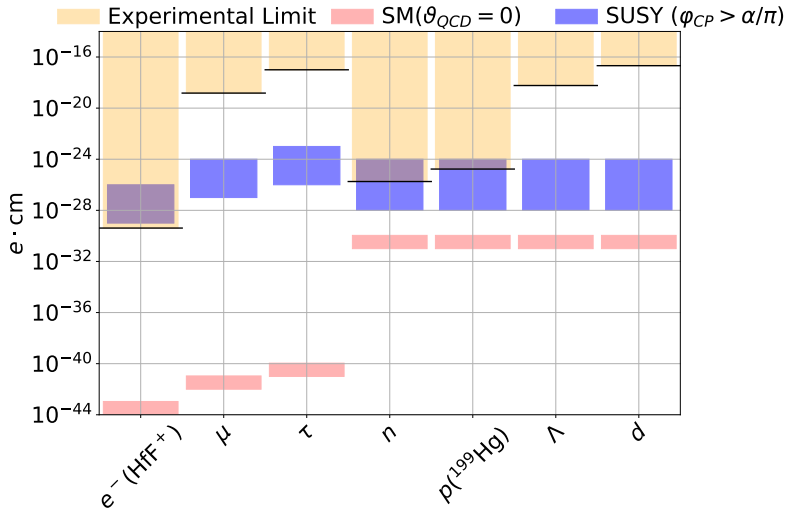
Summary & Outlook

Achievements at Cooler Synchrotron COSY at Forschungszentrum Jülich:

- large spin coherence time → observing 120kHz spin precession → feedback system on polarisation → build-up of vertical polarisation observed
- first measurement of deuteron EDM
- first search for axion-like particle in storage ring

- precision limited because of magnetic storage ring
- future: Dedicated storage ring needed

EDM limits



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